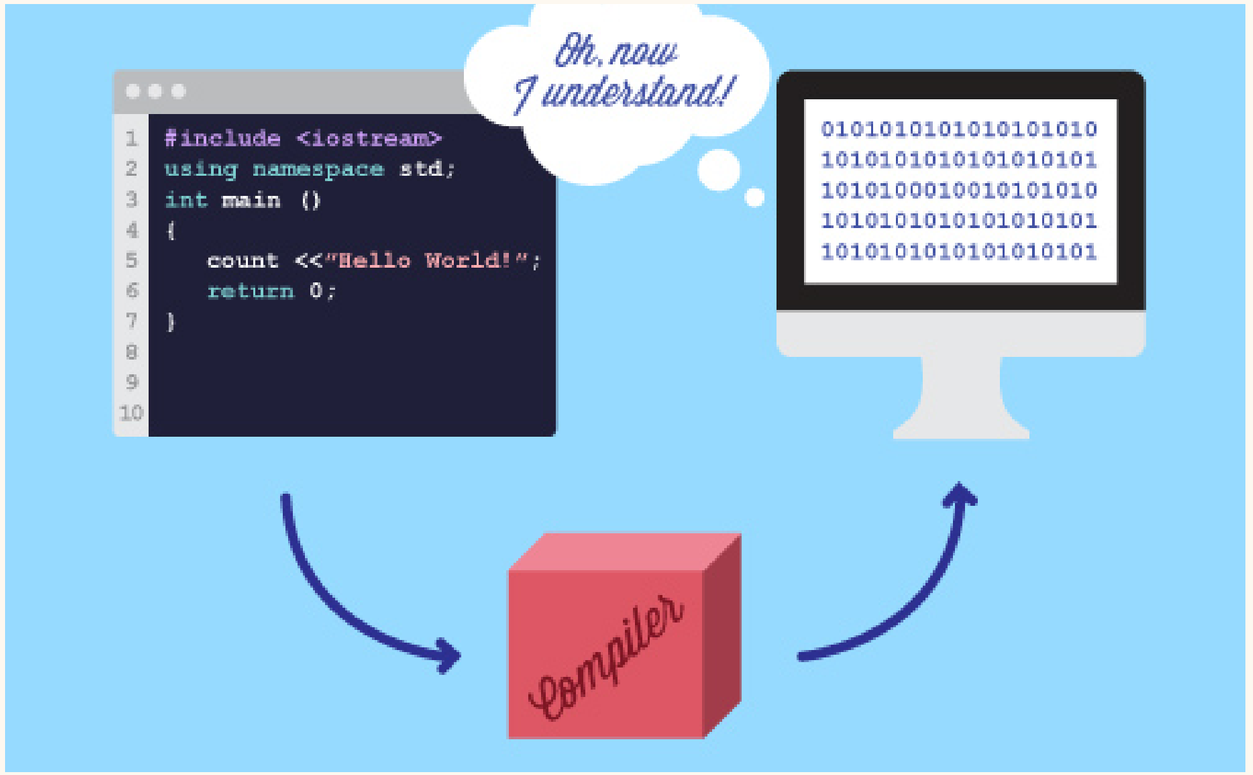
Parser & STD



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Compilers
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One syntax error can cost hours

The problem:

A missing “ ; ” or a wrong type, and the compiler message does not tell you where to look.

Our solution:

An analyzer that checks the code and reports the error with the exact line number.

Three stages, one clear result

1. Lexer

Splits the text and labels each piece: keyword, identifier, operator, constant, punctuation.

2. Parser

Checks the structure is correct using a left-factored LL(1) grammar.

3. SDT

Checks the structure is correct using a left-factored LL(1) grammar.

An example: int x = 10;

Lexeme	Category
int	Keyword
x	Identifier
=	Operator
10	Constant
;	Punctuation

Left factoring

int x; and int x = 10; share the prefix type + id. The Parser reads it once, then decides: ; or = value ;

Three cases

Case	Input	Result
Valid	int x = 10;	No errors
Syntactic	int x	Error in line 1
Semantic	int x = “hola”;	Type error line 1